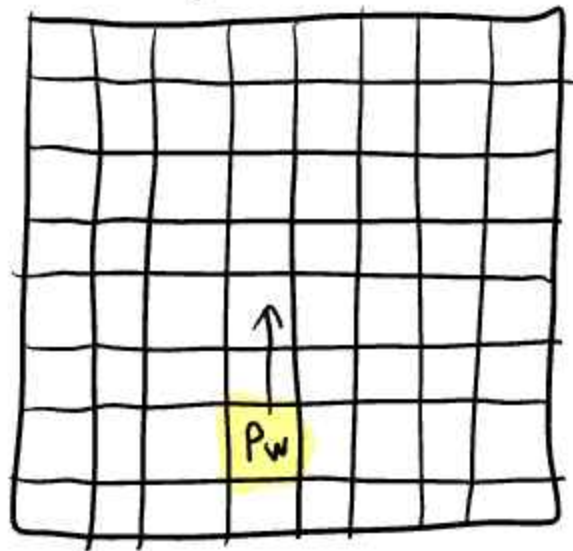


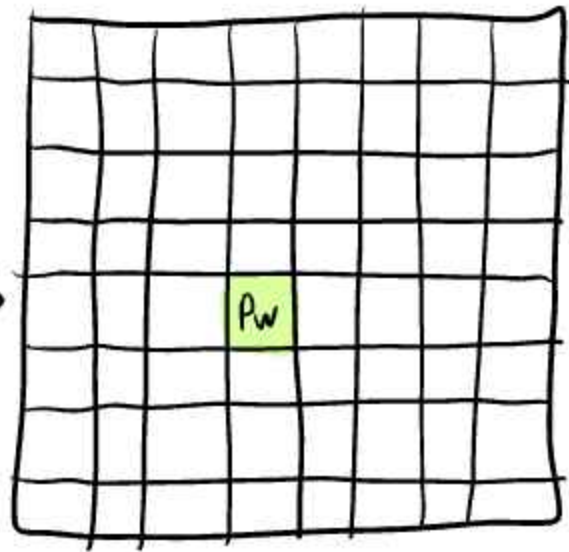
Term Project 1 Storyboard

Anthony Tom
Andrew ID: amtom

1) Moving Pieces

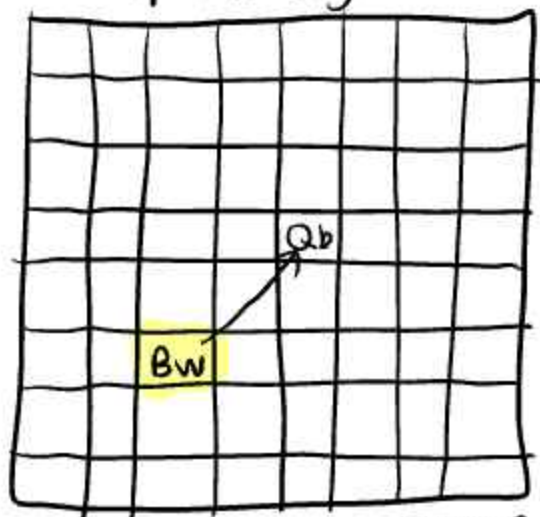


A player can click on a cell with a piece. The cell will be highlighted. This utilizes cell selection.

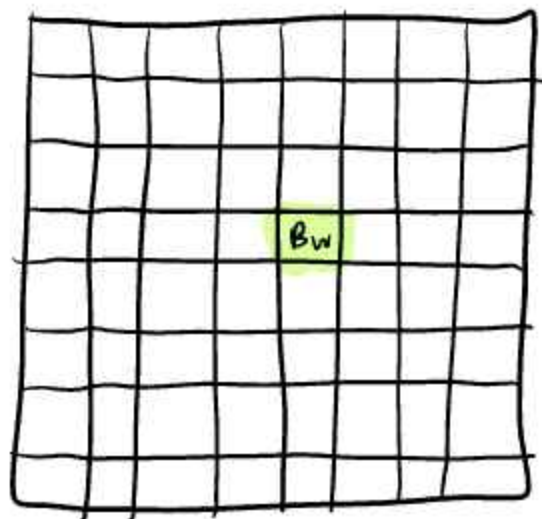


The player can click on another cell to move the chess piece if the move is valid.

2) Capturing Pieces

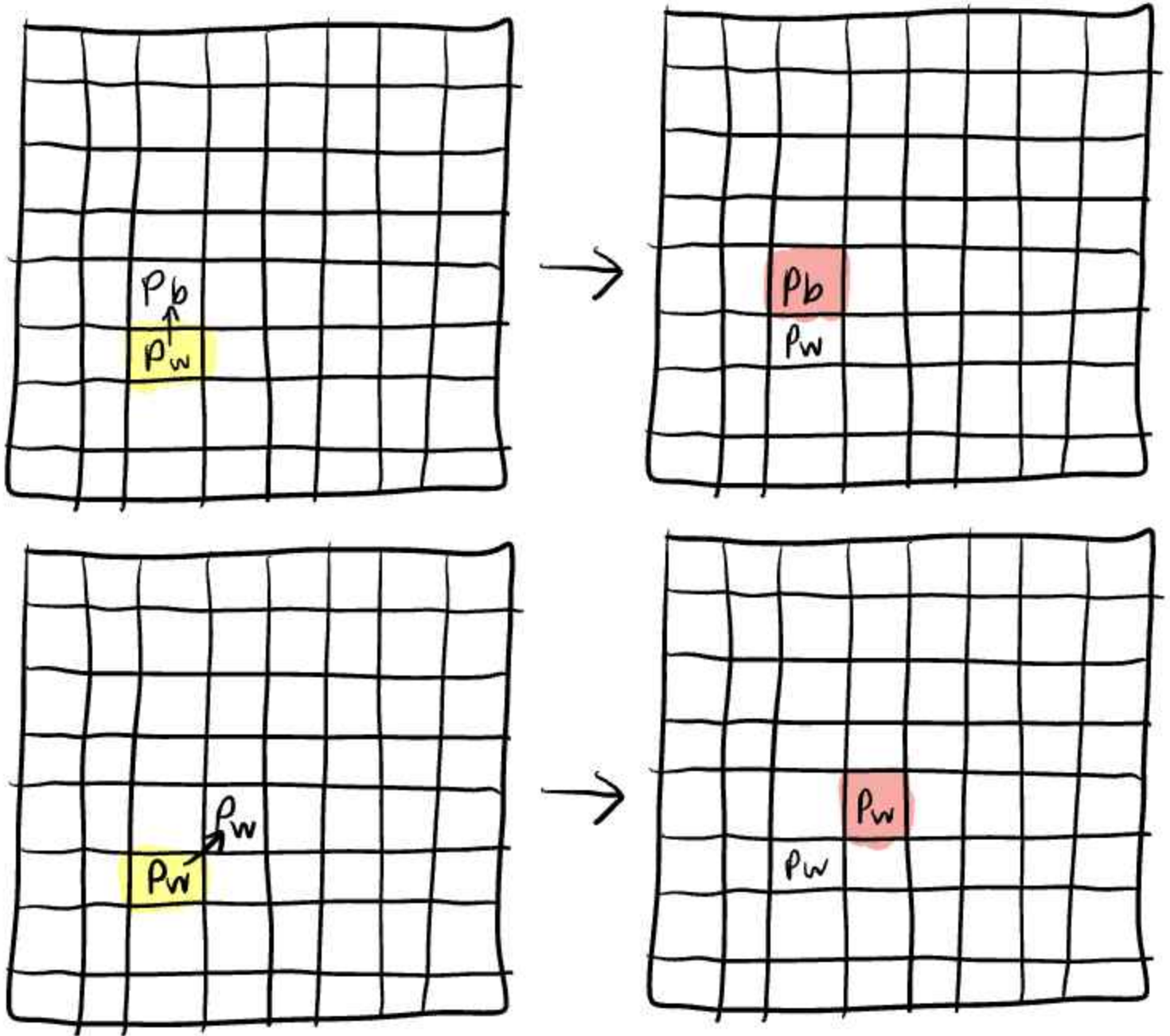


First, the player clicks on a piece to be moved.



If the other piece is a different color, the clicked piece replaces the other piece.

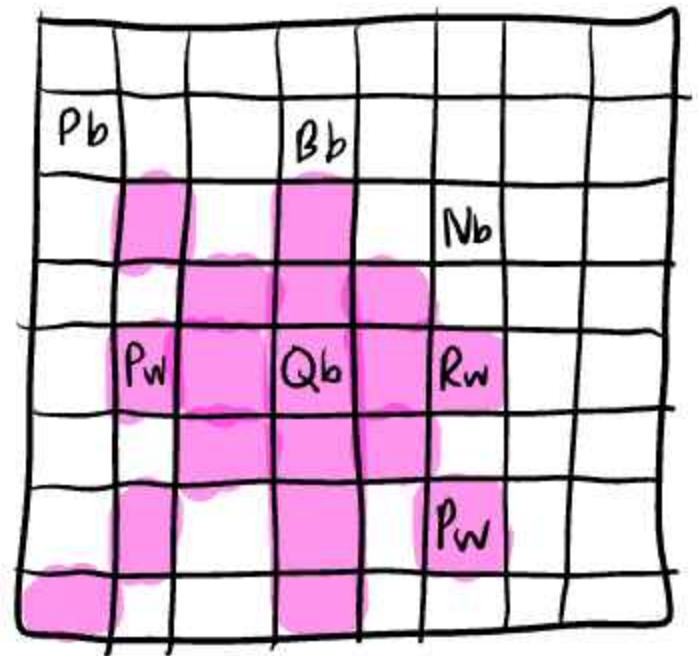
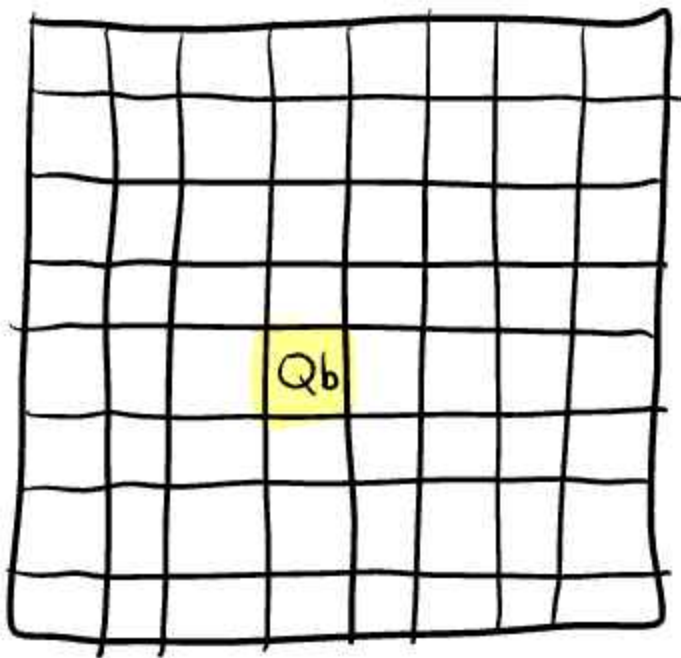
3) Invalid Moves



If a player tries to move a piece into an invalid cell, that cell is highlighted in red or another significant color.

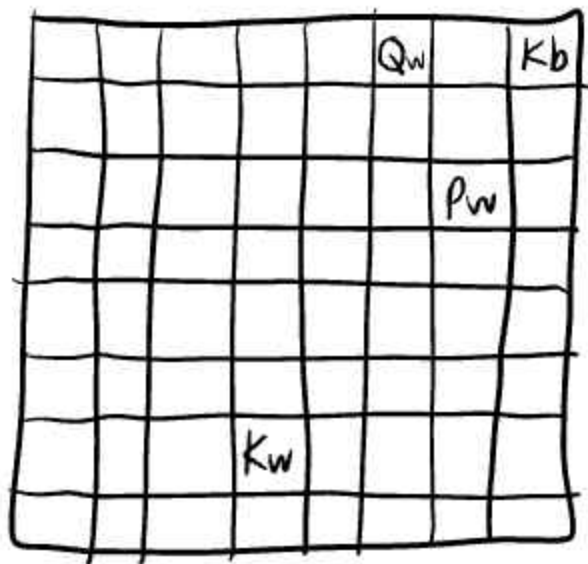
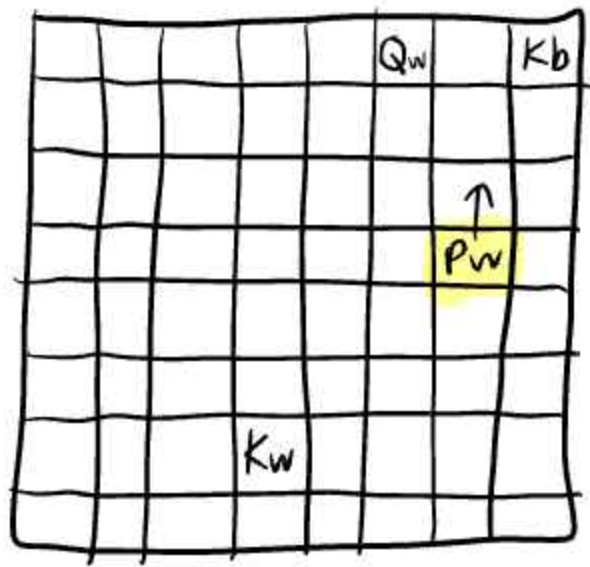
4) Possible Moves Detection

 Assistance
(toggle)



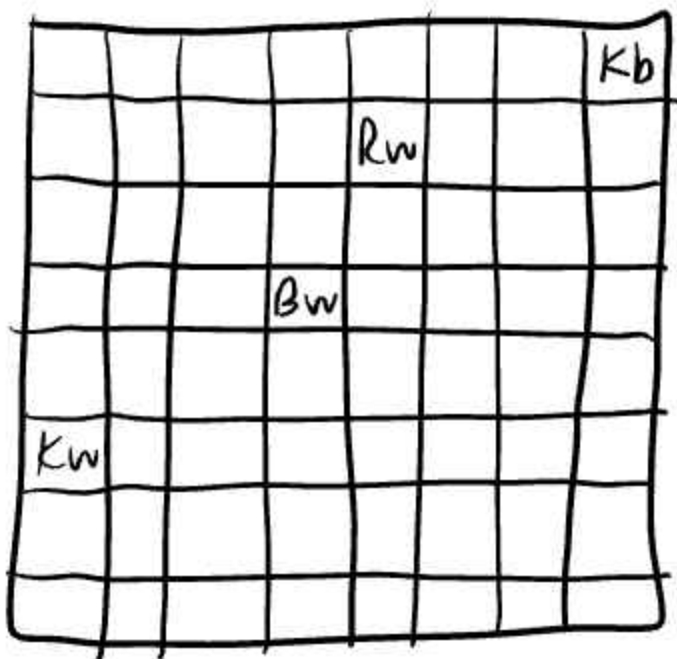
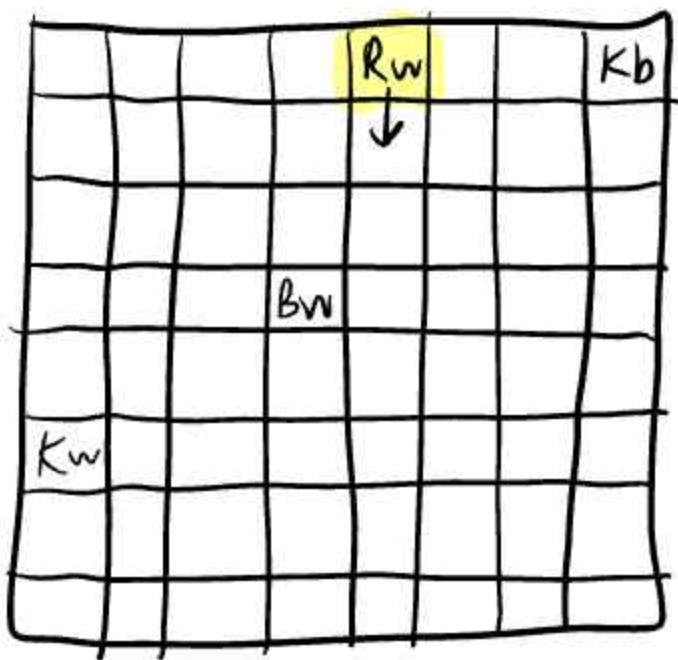
A player can toggle on "Assistance" to see the possible moves of chess pieces they click on (cells that the piece can be moved to are marked).

5) Stalemate, Checkmate



Player White
Wins by
Checkmate!

If the board is in checkmate position (the loser's king will be captured), a message is displayed indicating the winner and the checkmate condition.



Tie!
(Stalemate)

If the board is in stalemate position (one player has no legal moves), a message is displayed indicating the stalemate condition.

6) Chess Timers

Rb	Nb	Bb	Qb	Kb	Bb	Nb	Rb
Pb	Pb	Pb	Pb	Pb		Pb	Pb
					Pb		
				Pw			
Pw	Pw	Pw	Pw		Pw	Pw	Pw
Rw	Nw	Bw	Qw	Kw	Bw	Nw	Rw

1:12

1:28 (↓)

Using the built-in time module, chess timers can be added. When it is a player's turn, their timer is highlighted and decreases. When their timer reaches 0:00, the other player wins by time and a message appears displaying the winner that wins by time.

Example:



1:00

0:00



Player
wins by
time

7) Chess AI

AI

Rb	Nb	Bb	Qb	Kb	Bb		Rb
Pb	Pb	Pb	Pb	Pb		Pb	Pb
					Pb		Nb
				Pw			
			Pw				
Pw	Pw	Pw			Pw	Pw	Pw
Rw	Nw	Bw	Qw	Kw	Bw	Nw	Rw

Player

Using recursion and the minimax algorithm, a Chess AI player can be created. It would move pieces to cells on its own and analyze various chess positions to determine good moves to make.